END Term Examination – 2025

Name of the Course: M.Tech
Name of the Paper: Machine Learning
Time: 3 Hours

Semester: 1st
Paper Code: MCS 128
Maximum Marks: 100

Note:-
(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.

Q1 (20 marks)

a) Explain the role of data preprocessing in building a machine learning model. Discuss techniques for data cleaning, feature scaling, and encoding categorical data.

b) What are some common challenges or limitations of linear regression, and when might it not be the appropriate choice for a modeling task? Discuss with a suitable example. **CO1, CO2**

c) What is the difference between classification and regression in machine learning? Discuss with suitable examples.

Q2 (20 marks)

a) Explain the concept of classification and its types. Describe the difference between binary, multiclass, and multilabel classification problems with examples.

b) Explain the process of building a Decision Tree. Describe how Entropy and Information Gain are calculated with a small example. **CO2, CO3**

c) Explain the working principles of support vector machines (SVM) and when it is suitable for classification problems.

Q3 (20 marks)

a) Explain the concept of Hierarchical Clustering in unsupervised learning. Discuss how the dendrogram represents the hierarchy of clusters. Describe how the choice of linkage criteria (such as single, complete, or average linkage) affects the clustering results. Support your answer with a suitable example. **CO3, CO4**

b) Explain how the K-Means algorithm works step-by-step. Discuss the roles of initial centroids, distance calculation, and cluster updates.

c) The DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm is often used to discover clusters in spatial and high-dimensional data. Determine the following:
(i) Explain the working principle of DBSCAN and describe how it distinguishes core, border, and noise points.
(ii) Consider a scenario where a researcher is analyzing GPS data of vehicles in a city to identify traffic hotspots and outlier movements. Discuss how DBSCAN can be effectively applied in this context.
(iii) Compare its performance with K-Means for such spatial data and state one limitation of DBSCAN.

Q4 (20 marks)

a) Explain the working of a Artificial Neural Network (ANN). Describe the role of activation functions, weights, and biases in its operation.

b) Consider a perceptron with inputs $x_1 = 1$, $x_2 = -1$, weights $w_1 = 0.8$, $w_2 = 0.4$, and bias $b = -0.2$. The activation function outputs 1 if the net input ≥ 0, otherwise 0. Compute the perceptron's final output. **CO4, CO5**

c) Define Reinforcement Learning. Explain the roles of agent, environment, states, actions, and rewards in an RL system using a clear diagram.

Q5 (20 marks)

a) Explain the concept of gradient-based optimization. Discuss the differences between Batch Gradient Descent, Stochastic Gradient Descent (SGD), and Mini-batch Gradient Descent.

b) Explain the architecture of a Convolutional Neural Network (CNN) in detail. Describe the roles of convolution layers, pooling layers, and fully connected layers. **CO5, CO6**

c) Compare RNN, LSTM, and GRU in terms of structure, training complexity, and memory retention ability. Provide a suitable example for each.